

Primary 4-6

12-WEEK TEACHER LESSON PLAN

QUESTION -> MEASURE -> INVESTIGATE -> ANALYZE -> EXPLAIN

Structure	Weeks 1-10: one GrowMeal garden per week. Week 11: synthesis, remediation and portfolio. Week 12: celebration, demonstration and level completion.
Integrated resources	Vocabulary -> Flashcards -> Teacher Demonstration -> Practical Activity -> Journal -> Garden Mission -> Ingredient Lab/Farm-to-Food/Food Smart -> Assessment -> Passport.
Teacher preparation	Inspect the garden/site, choose exact cards/activities, prepare materials and DIY substitutes, review safety, assign learner groups and pre-check instruments.
Evidence rule	Every week should leave visible evidence: oral/drawn evidence at Nursery; records and measurements in Primary; increasingly traceable data, analysis and decisions at JSS/SSS.
Safety	School safeguarding, food-safety and tool-use rules take priority. No tasting is required. Teachers control knives, heat, chemicals, powered equipment and other higher-risk tools.

WEEK 01 - Raised Food Garden

Learning objective	Frame and investigate a measurable question in the Raised Food Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Raised Food Garden; TD-201 to TD-210; PA-201 to PA-210; GM-201 to GM-210; AQ-201 to AQ-210.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-201–IL-210, FF-201–FF-210, or FSC-201–FSC-210 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-201–AQ-210; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 02 - Geometry Garden

Learning objective	Frame and investigate a measurable question in the Geometry Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Geometry Garden; TD-211 to TD-220; PA-211 to PA-220; GM-211 to GM-220; AQ-211 to AQ-220.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-211–IL-220, FF-211–FF-220, or FSC-211–FSC-220 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-211–AQ-220; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 03 - Grid Experiment Garden

Learning objective	Frame and investigate a measurable question in the Grid Experiment Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Grid Experiment Garden; TD-221 to TD-230; PA-221 to PA-230; GM-221 to GM-230; AQ-221 to AQ-230.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-221–IL-230, FF-221–FF-230, or FSC-221–FSC-230 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-221–AQ-230; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 04 - Keyhole Garden

Learning objective	Frame and investigate a measurable question in the Keyhole Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Keyhole Garden; TD-231 to TD-240; PA-231 to PA-240; GM-231 to GM-240; AQ-231 to AQ-240.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-231–IL-240, FF-231–FF-240, or FSC-231–FSC-240 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-231–AQ-240; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 05 - Compost Garden

Learning objective	Frame and investigate a measurable question in the Compost Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Compost Garden; TD-241 to TD-250; PA-241 to PA-250; GM-241 to GM-250; AQ-241 to AQ-250.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-241–IL-250, FF-241–FF-250, or FSC-241–FSC-250 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-241–AQ-250; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 06 - Gravity Drip Garden

Learning objective	Frame and investigate a measurable question in the Gravity Drip Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Gravity Drip Garden; TD-251 to TD-260; PA-251 to PA-260; GM-251 to GM-260; AQ-251 to AQ-260.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-251–IL-260, FF-251–FF-260, or FSC-251–FSC-260 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-251–AQ-260; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 07 - Vertical Garden

Learning objective	Frame and investigate a measurable question in the Vertical Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Vertical Garden; TD-261 to TD-270; PA-261 to PA-270; GM-261 to GM-270; AQ-261 to AQ-270.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-261–IL-270, FF-261–FF-270, or FSC-261–FSC-270 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-261–AQ-270; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 08 - Crop Rotation Garden

Learning objective	Frame and investigate a measurable question in the Crop Rotation Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Crop Rotation Garden; TD-271 to TD-280; PA-271 to PA-280; GM-271 to GM-280; AQ-271 to AQ-280.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-271–IL-280, FF-271–FF-280, or FSC-271–FSC-280 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-271–AQ-280; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 09 - Water-Smart Garden

Learning objective	Frame and investigate a measurable question in the Water-Smart Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Water-Smart Garden; TD-281 to TD-290; PA-281 to PA-290; GM-281 to GM-290; AQ-281 to AQ-290.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-281–IL-290, FF-281–FF-290, or FSC-281–FSC-290 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-281–AQ-290; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 10 - Intro Soilless Garden

Learning objective	Frame and investigate a measurable question in the Intro Soilless Garden, analyze evidence and explain a limitation.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Flashcards for Intro Soilless Garden; TD-291 to TD-300; PA-291 to PA-300; GM-291 to GM-300; AQ-291 to AQ-300.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-291–IL-300, FF-291–FF-300, or FSC-291–FSC-300 activity where it fits the week's garden evidence.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Use observation plus AQ-291–AQ-300; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 11 - Synthesis & Portfolio

Learning objective	Synthesize evidence from the ten gardens, revisit weak competencies, complete unfinished missions, and prepare a portfolio of the learner's strongest evidence.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Student Journal; Food Smart Passport; selected Practical Cards, Garden Missions, Ingredient Lab, Farm-to-Food and Food Smart Challenge evidence from Weeks 1-10.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Learners choose evidence that shows growth in knowledge, practical competence, investigation/data skills and mission-based problem solving.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Teacher moderation: review journal, passport, assessment results and practical evidence; reteach or reassess competencies not yet secure.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 12 - GrowMeal Celebration & Demonstration

Learning objective	Celebrate learning through a safe GrowMeal showcase in which learners communicate what they grew, observed, measured, investigated and solved.
Vocabulary	10-15 scientific terms linked to variables, measurement, data, garden processes and evidence.
Resources	Food Smart Passport; learner portfolio; garden displays; selected charts/models; certificates or school-approved recognition materials.
Materials / DIY	String sampling grids, marked containers, handmade stakes, graph paper and simple locally available measuring tools.
Suggested lesson flow	Investigation question (10 min) -> method/safety demonstration (10 min) -> sampling/data collection (30 min) -> calculation/graph (15 min) -> conclusion/mission (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Garden walk, learner demonstrations, evidence exhibition, Farm-to-Food/Ingredient Lab displays and age-appropriate challenge presentations.
Questions to ask	Is the comparison fair? What changes? What stays constant? What pattern appears? What limitation matters?
Assessment	Final teacher verification, learner reflection, level-completion record and next-level learning goals.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?